

and note that the origin of time is irrelevant, then the equation for the evolution of  $\phi$  could be written

$$\frac{d}{dt}\hat{\phi} = (\mathcal{L}_f\phi)(z). \quad (3.2)$$

Similarly,

$$\frac{d^2}{dt^2}\phi(z(t)) = (\mathcal{L}_f^2\phi)(z(t)).$$

The Taylor series expansion of  $\phi(z(t))$  along a solution of the differential equation can therefore be written as

$$\begin{aligned} \phi(z(t)) &= \phi(z(0)) + t \left. \frac{d}{dt}\phi(z(t)) \right|_{t=0} + \frac{t^2}{2} \left. \frac{d^2}{dt^2}\phi(z(t)) \right|_{t=0} + \dots \\ &= \phi(z(0)) + t(\mathcal{L}_f\phi)(z(0)) + \frac{t^2}{2}(\mathcal{L}_f^2\phi)(z(0)) + \dots \\ &= (e^{t\mathcal{L}_f}\phi)(z(0)). \end{aligned}$$

This gives a concise formula for the evolution of any function of the phase variables, including, in particular, any solution component  $z_i$ . Thus the flow map of the system can be represented by  $\exp(t\mathcal{L}_f)$ . When one writes  $\mathcal{F}_t = \exp(t\mathcal{L}_f)$ , what is actually meant is that the individual components satisfy

$$z_i(t, \xi) = [\mathcal{F}_t(\xi)]_i = (\exp(t\mathcal{L}_f)z_i)|_{z=\xi}.$$

By analogy with the derivation of the matrix exponential, it is tempting to express the exponential as a series expansion in powers of  $t$ :

$$e^{t\mathcal{L}_f} = \text{Id} + t\mathcal{L}_f + \frac{t^2}{2}\mathcal{L}_f^2 + \dots$$

However, as  $\mathcal{L}_f$  is a first order operator, this series involves high order mixed partial derivatives, and the convergence of the expansion has to be considered in connection with its application to given initial data, that is, we should consider the expansion

$$e^{t\mathcal{L}_f}\phi = \phi + t\mathcal{L}_f\phi + \frac{t^2}{2}\mathcal{L}_f^2\phi + \dots$$

then the primary issue is the boundedness of the terms with respect to functions  $\phi$  of a certain space of functions; we ignore the convergence issue here and treat the operator exponential as a formal series expansion. Methods derived using the formal expansion technique should be subject to further investigation (either numerical or analytical) to establish their validity.